

HW 0303

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1.

(a)

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

(b)

$$b_2 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{8}{10} = 0.8,$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 * 1 = 1.2$$

(c)

$$\sum_{i=1}^5 x_i^2 = 15,$$

$$\sum_{i=1}^5 x_i y_i = 18,$$

$$\sum_{i=1}^N (x_i - \bar{x})^2 = 10,$$

$$\sum_{i=1}^N x_i^2 - N \bar{x}^2 = 15 - 5 * 1 = 10,$$

$$\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) = 8,$$

$$\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y} = 18 - 5 * 1 * 2 = 8$$

$$\Rightarrow \sum_{i=1}^N (x_i - \bar{x})^2 = \sum_{i=1}^N x_i^2 - N \bar{x}^2,$$

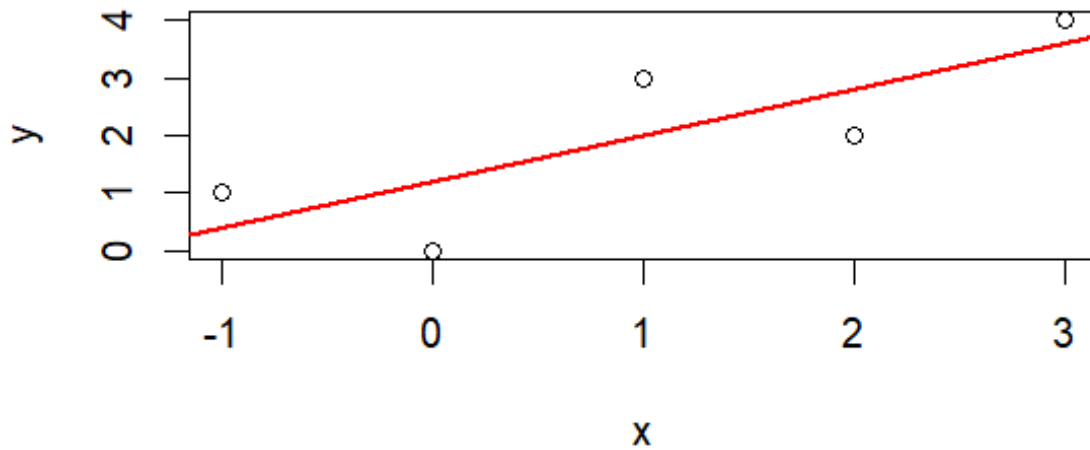
$$\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}$$

(d)

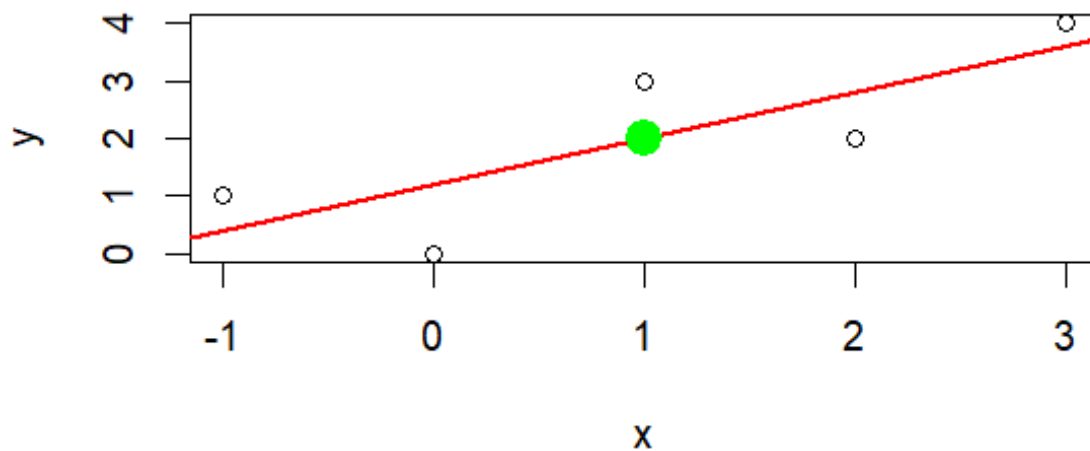
x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2.0	1.0	1.00	1.0
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0.0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$\begin{aligned}s_x^2 &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{10}{5-1} = 2.5, \\s_y^2 &= \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1} = \frac{10}{5-1} = 2.5, \\s_{xy} &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{8}{5-1} = 2, \\r_{xy} &= \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5} * \sqrt{2.5}} = 0.8 \\CV_x &= 100 \left(\frac{s_x}{\bar{x}} \right) = 100 * \frac{\sqrt{2.5}}{1} = 158.11\end{aligned}$$

(e)



(f)



Fitted line pass through the point.

(g)

$$\begin{aligned}\bar{y} &= 2, \\ b_1 + b_2\bar{x} &= 1.2 + 0.8 * 1 = 2, \\ \Rightarrow \bar{y} &= b_1 + b_2\bar{x}\end{aligned}$$

(h)

$$\hat{\bar{y}} = 10/5 = 2 = \bar{y}$$

(i)

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^N \hat{e}_i^2}{N - k} = \frac{3.6}{5 - 2} = 1.2$$

(j)

$$\begin{aligned}\text{Var}(b_2|X) &= \frac{\hat{\sigma}^2}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12 \\ \text{se}(b_2) &= \sqrt{\text{Var}(\hat{\beta}_2|X)} = \sqrt{0.12} = 0.35\end{aligned}$$

22.

(a)

$$TOTALSCORE_i = 916.442 + 12.175SMALL_i$$

Since $b_2 = 12.175 > 0$, we find that students in small classes perform better than those in regular-sized classes.

(b)

$$READSCORE_i = 432.665 + 6.924SMALL_i$$

$$MATHSCORE_i = 483.7767 + 5.251SMALL_i$$

$432.665 + 483.7767 \approx 916.442$, $6.924 + 5.251 \approx 12.175$. Comparing the impact of class size, we observe that the positive effect is more substantial on READSCORE than on MATHSCORE ($6.924 > 5.251$), indicating that small classes contribute more to reading proficiency than to mathematical performance.

(c)

$$TOTALSCORE_i = 916.442 + 4.306SMALL_i$$

Since $b_2 = 4.306 > 0$, we find that students in classrooms with a teacher aide perform better than those in classrooms without one.

(d)

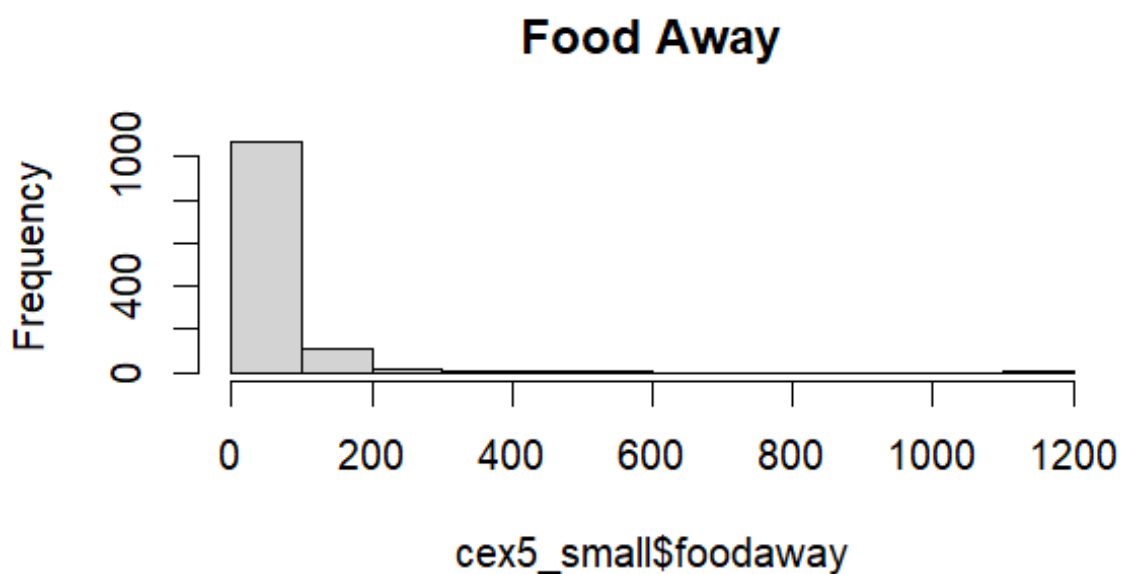
$$READSCORE_i = 432.665 + 2.871SMALL_i$$

$$MATHSCORE_i = 483.777 + 1.435SMALL_i$$

Comparing the models, we find that the coefficient for SMALL is substantially larger than that for AIDE across all subjects. Specifically, the impact of AIDE is more pronounced on READSCORE than on MATHSCORE, too.

25.

(a)



mean : 49.27
 median : 32.55
 25th percentiles : 12.04
 75th percentiles : 67.50

(b)

With a member with an advanced degree:

mean : 73.15, median : 48.15

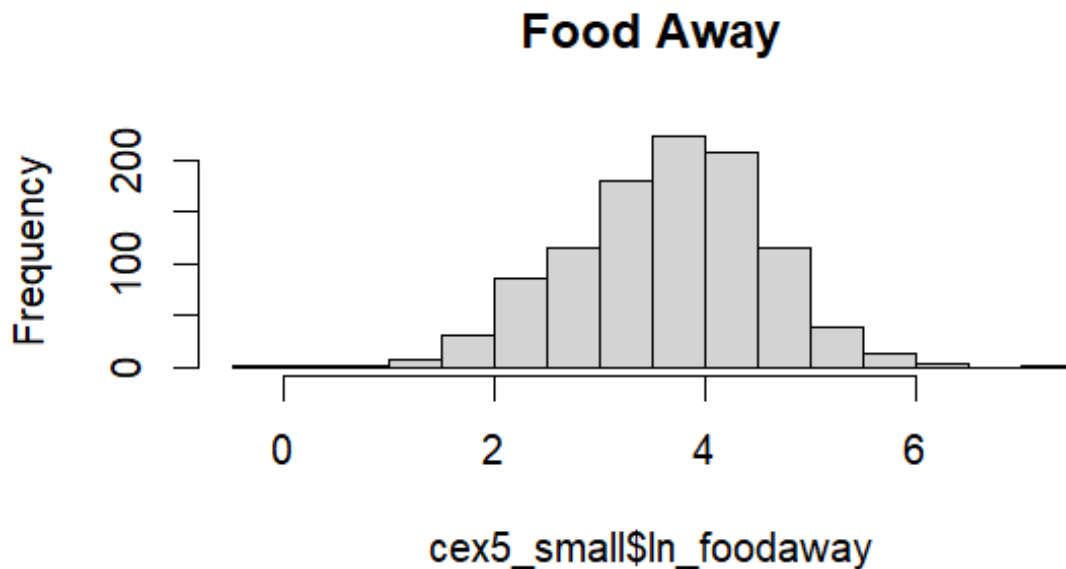
With a college degree member:

mean : 48.60, median : 36.11

With no advanced or college degree member:

mean : 39.01, median : 26.02

(c)



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	9.63	26.02	39.01	52.65	437.78

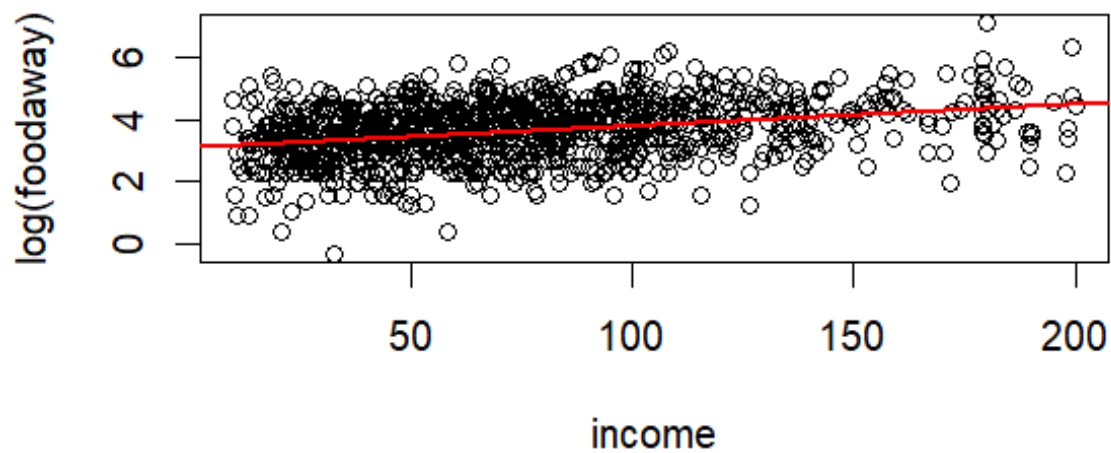
Since $\ln(x)$ is undefined for non-positive values. All observations where FOODAWAY equals zero are excluded from the sample during the log transformation. Resulting in different numbers of observations.

(d)

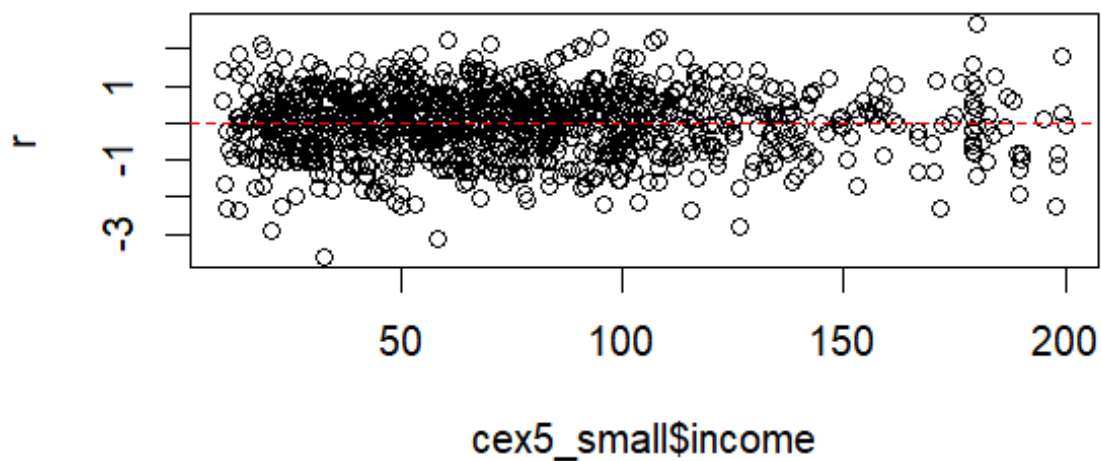
$$\ln(FOODAWAY) = 3.1293 + 0.0069INCOME$$

$b_2 = 0.0069$ implies that for every one-unit increase in income, the average expenditure on food away from home is expected to increase by 0.69%

(e)



(f)



The residual plot displays a random distribution around the horizontal axis, indicating no significant non-linear patterns. The data points are densely concentrated within the $\text{INCOME} < 150$ range but no discernible pattern of expansion or contraction as income increases. They seem completely random.